
DISCOVERY PHASE ARTIFACT PACK

Customer Health Discovery and Solution Design

Business canvas, process flow, architecture, integrations, swimlanes, application design, PRD, and AI agent graph for the Customer Health project.

PROJECT

Customer Health

PHASE

Discovery

GENERATED

May 20, 2026

This artifact pack captures the discovery and solution design baseline for the Customer Health project. It is intended to be reviewed during Discovery, refined with stakeholders, and used as the input for implementation planning.

Artifact Register

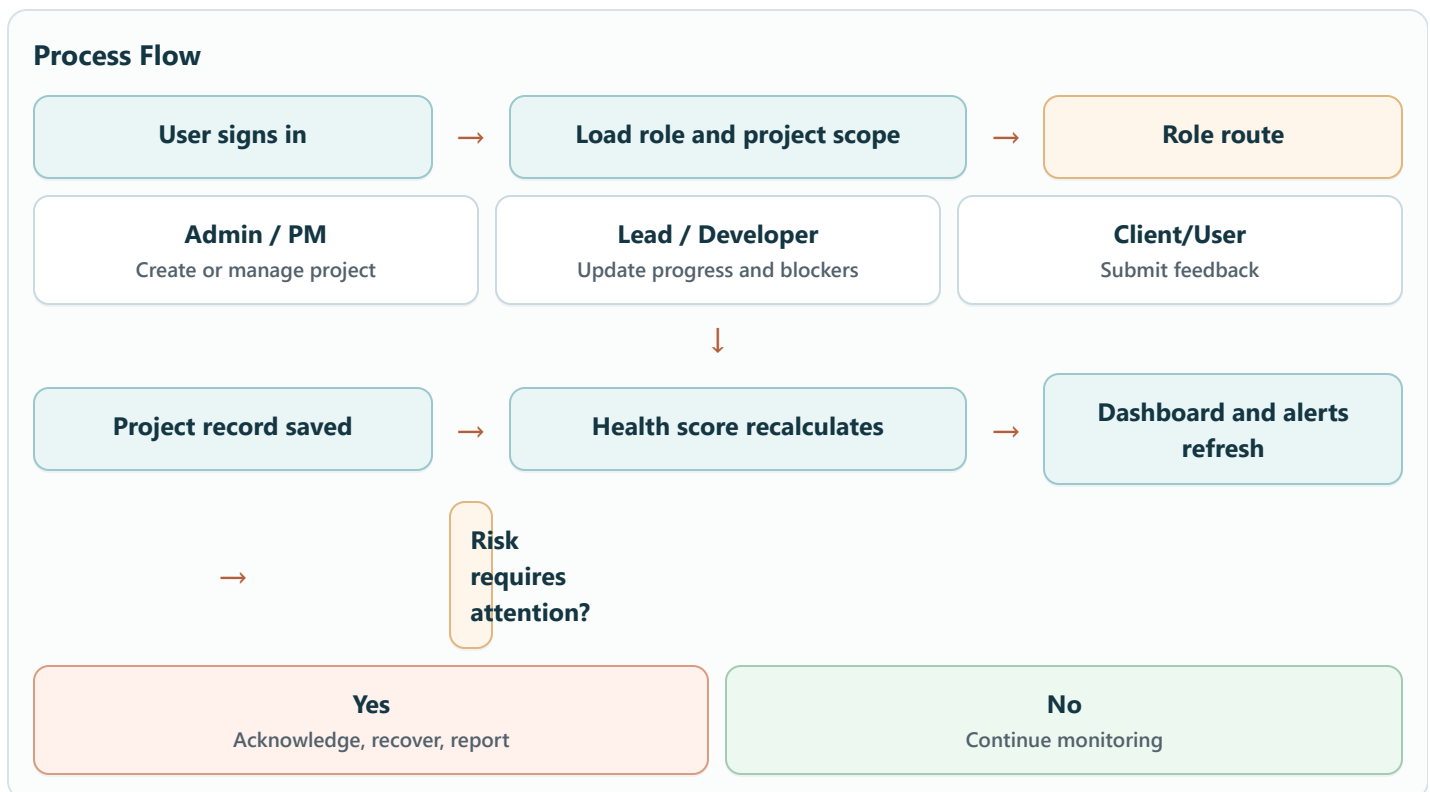
Artifact	Purpose	Discovery Output
Business Canvas	Align business problem, value, users, and success metrics.	Shared view of why Customer Health exists and how value is measured.
Process Flow	Map the core end-to-end operating process.	Intake-to-escalation workflow for customer project health monitoring.
Functional Architecture Diagram	Define major application capabilities and boundaries.	Logical architecture for UI, API, roles, health scoring, alerts, and reporting.
Integration Diagram	Show internal and external system touchpoints.	Current and target integrations across identity, delivery tools, CRM, support, and analytics.
Swimlane Diagram	Clarify responsibilities across roles.	Role-based workflow across Admin, Project Manager, Team Lead, Developer, Client/User, and System.
Application Design	Describe screens, modules, data objects, and role behavior.	Product design baseline for the current app and near-term enhancements.
PRD	Convert discovery findings into product requirements.	Goals, users, requirements, acceptance criteria, constraints, and rollout plan.
Agent Graph	Define AI-assisted monitoring agents and human review loops.	Agent workflow for ingestion, scoring, risk detection, recommendations, and reporting.

1. Business Canvas

Area	Discovery Notes
Business problem	Delivery leaders need an early-warning system for customer project health, escalation risk, and role-based project visibility. Today, risk signals can be scattered across project updates, client feedback, technical blockers, and manager status reports.
Target users	Admins, Project Managers, Team Leads, Developers, and Client/Users.
Current customer examples	Signal, REI Blackbook, and Cafe Zupas are represented as seeded customer delivery records.
Value proposition	Provide one governed workspace for project health, weighted health scoring, risk alerts, client feedback, accountable ownership, and exportable delivery reporting.
Key workflows	Login and role scoping, project creation, project health updates, client satisfaction feedback, risk acknowledgement, portfolio dashboarding, and report export.
Differentiators	Health score combines progress, team performance, client satisfaction, delivery confidence, and explicit risk level penalties. Visibility is scoped by actor role and assigned project.

Area	Discovery Notes
Success metrics	Reduced unresolved escalations, faster risk acknowledgement, improved delivery confidence, increased client satisfaction, on-time milestone recovery, and adoption by project roles.
Business constraints	Customer data must be access controlled. Client users should see only client-facing risk details. Delivery teams need low-friction updates rather than a heavy project management tool.
Assumptions	Project health can be improved when risk signals are visible early, assigned to accountable owners, and translated into next actions.
Open discovery questions	Which upstream tools are the source of truth for projects, tasks, support tickets, and client sentiment? Which score thresholds trigger required escalation? What audit trail is required for regulated customers?

2. Process Flow



Process Notes

1. Access is role-based and project-scoped before any project data is shown.
2. Project updates can change progress, status, risk level, team performance, client satisfaction, and delivery confidence.
3. Client feedback can add client-visible risk context and influence the health score.
4. Risk acknowledgement is limited to Admin and Project Manager roles.
5. Reports are generated from the current scoped application state.

3. Functional Architecture Diagram

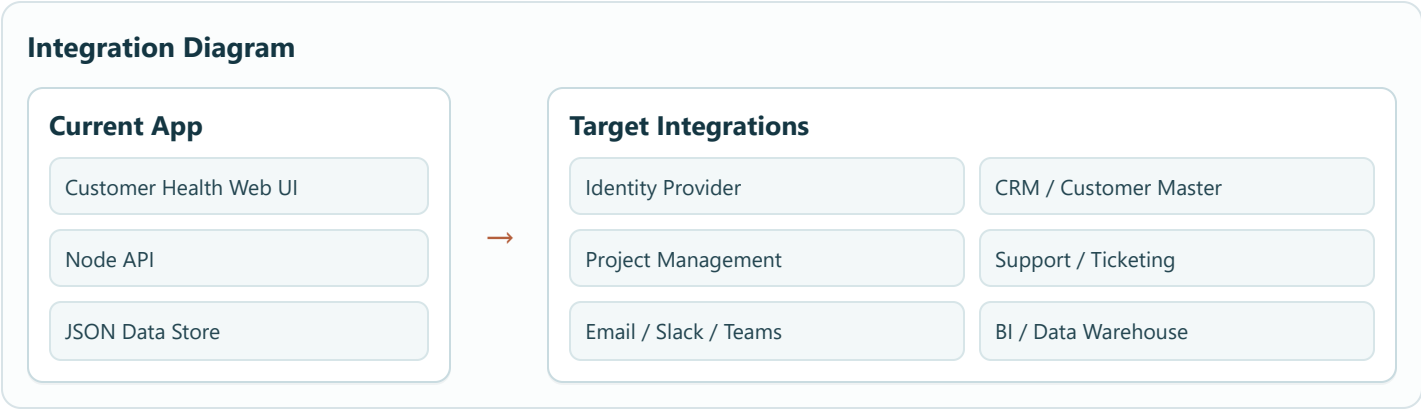
Functional Architecture

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Functional Capabilities

Capability	Current Baseline
Authentication	Local login and registration through backend API.
Authorization	Role and project assignment checks for viewing, editing, creating, acknowledging, and exporting.
Project health management	Project create/update, customer profile, progress, status, risk level, issue notes, and next actions.
Client feedback	Client/User satisfaction and client-visible risk feedback.
Health scoring	Weighted score using progress, team performance, client satisfaction, delivery confidence, and risk penalty.
Risk center	Filters and alerts for projects requiring attention.
Reporting	Scoped export endpoint and in-app forecast/report views.
Deployment	Node server with production/staging environment examples and static build support.

4. Integration Diagram

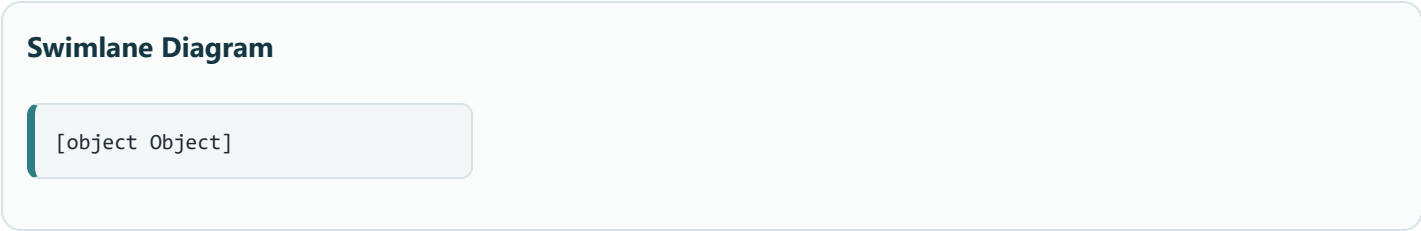


Integration Scope

Integration	Direction	Discovery Decision Needed
Identity provider	Inbound	Confirm SSO provider, role claims, session duration, and group-to-role mapping.

Integration	Direction	Discovery Decision Needed
CRM/customer master	Inbound	Confirm source of customer accounts, renewal dates, account owners, and contract metadata.
Project management	Inbound	Confirm source for milestones, tasks, blockers, sprint progress, and ownership.
Support/ticketing	Inbound	Confirm incident severity, SLA status, defects, and support response time fields.
Communications	Outbound	Confirm alert channels, escalation recipients, digest cadence, and opt-out rules.
BI/data warehouse	Outbound	Confirm reporting granularity, retention, audit requirements, and schema ownership.

5. Swimlane Diagram



Role Responsibilities

Role	Responsibilities
Admin	Own system configuration, access, governance, portfolio reporting, and executive visibility.
Project Manager	Own customer delivery health, risk response, stakeholder updates, timeline, and forecast.
Team Lead	Update team progress, delivery blockers, implementation risks, and technical readiness.
Developer	Update assigned implementation progress and add issue notes.
Client/User	Review project summary, timeline, client-visible risks, and submit satisfaction feedback.
System	Enforce access, calculate health, surface alerts, refresh dashboards, and prepare reports.

6. Application Design

Information Architecture

Area	Design Intent
Login and registration	Authenticate users and assign scoped access to the selected project.
Dashboard	Provide portfolio-level health, risk, delivery, activity, and prediction summaries.

Area	Design Intent
Projects	List customer records with filtering, selected profile details, and create/update workflows.
Health Metrics	Explain score drivers for the selected project.
Risk Alerts	Highlight projects needing attention and enable acknowledgement where permitted.
Access Control	Show actors, responsibilities, and role permissions.
Reports	Provide delivery forecast, status summaries, brief lists, and export action.
Proposal	Explain escalation prevention drivers and workflow-specific risk signals.

Data Model

Object	Key Fields
Actor	id, email, password, name, role, organization, projectIds, accessLevel, responsibilities.
Project	id, name, customer, status, health, progress, teamPerformance, clientSatisfaction, deliveryConfidence, riskLevel, environment, portfolioShare.
Timeline	start, milestone, delivery.
Ownership	projectManager, teamLead, developers, clientUser.
Risk model	risks, clientRisks, issues, nextActions, healthTrend, lastUpdated.
Acknowledgement	actor id mapped to acknowledged project ids.
Session	token mapped to actor id and expiration.

Health Score Design

The current scoring formula:

[object Object]

Risk penalty:

Risk Level	Penalty
Contained	0
Elevated	6
Critical	12

Role-Based Behavior

Action	Admin	Project Manager	Team Lead	Developer	Client/User
View all projects	Yes	Assigned only	Assigned only	Assigned only	Own project only
Create project	Yes	Yes	No	No	No
Edit project health	Yes	Assigned only	Assigned only	Assigned only	No

Action	Admin	Project Manager	Team Lead	Developer	Client/User
Submit satisfaction feedback	No	No	No	No	Own project only
Acknowledge risks	Yes	Assigned only	No	No	No
Export reports	Yes	Assigned only	Assigned only where allowed	No	No

7. Product Requirements Document

Product Summary

Customer Health is a role-based web application for monitoring customer delivery health, detecting escalation risk, and coordinating recovery actions across delivery, engineering, and client stakeholders.

Goals

1. Provide a single source of visibility for customer project health.
2. Detect delivery and satisfaction risks before they become escalations.
3. Give each role only the records and actions they are allowed to access.
4. Translate risk signals into accountable actions and exportable reporting.
5. Establish a foundation for AI-assisted project health monitoring.

Non-Goals

1. Replace the system of record for project plans or support tickets in the first release.
2. Automate irreversible escalation decisions without human review.
3. Expose internal engineering notes directly to client users.
4. Implement full enterprise SSO, data warehouse sync, or CRM sync before integration discovery is complete.

Personas

Persona	Need
Executive/Admin	Portfolio visibility, governance, access control, and exportable reports.
Project Manager	Current project health, risks, client sentiment, ownership, and next actions.
Team Lead	Clear view of blockers, team performance, and delivery confidence.
Developer	Simple workflow to add progress and issue notes.
Client/User	Trusted view of project status and a feedback channel for visible concerns.

Functional Requirements

ID	Requirement	Priority
FR-01	Users can log in and receive role-scoped application state.	Must
FR-02	Admins and Project Managers can create customer project records.	Must

ID	Requirement	Priority
FR-03	Authorized delivery users can update project status, risk level, progress, team performance, delivery confidence, and issue notes.	Must
FR-04	Client/Users can submit satisfaction feedback for their own project.	Must
FR-05	The system recalculates project health after project updates and client feedback.	Must
FR-06	The dashboard shows portfolio metrics, visible health, elevated risks, delivery confidence, and activity.	Must
FR-07	Risk alerts show projects requiring attention and allow authorized acknowledgement.	Must
FR-08	Access control views show role definitions, actors, permissions, and responsibilities.	Should
FR-09	Users with export permission can prepare delivery reports for their scoped project set.	Should
FR-10	The system can later ingest project, support, CRM, and communication signals from external integrations.	Should
FR-11	The system can generate AI-assisted risk summaries and next action recommendations with human review.	Could

Non-Functional Requirements

ID	Requirement
NFR-01	Access control must prevent users from viewing or changing unassigned project data.
NFR-02	Session handling must use secure cookies in production.
NFR-03	Project health recalculation must be deterministic and explainable.
NFR-04	The UI must remain usable on desktop and mobile viewports.
NFR-05	Deployment must support staging and production domains with HTTPS.
NFR-06	Future integrations must preserve auditability of imported signals and score changes.

Acceptance Criteria

Scenario	Acceptance Criteria
Role-scoped login	A Project Manager sees only assigned projects; an Admin sees all projects; a Client/User sees only their own project.
Health update	Updating progress or risk level changes the visible score and dashboard state without requiring a page reload.

Scenario	Acceptance Criteria
Client feedback	Client satisfaction feedback updates the project record and adjusts health scoring.
Risk acknowledgement	Only Admins and Project Managers can acknowledge visible risk alerts.
Report export	Export action returns a scoped report preparation response for the signed-in actor.
Deployment	/api/health returns an operational status from the deployed environment.

Release Plan

Phase	Scope
Discovery	Confirm personas, workflows, integrations, scoring model, data sensitivity, and reporting needs.
MVP	Role-scoped dashboard, project health records, health scoring, risk alerts, feedback, and reports.
Integration Pilot	Add one source integration for project status and one source integration for support or incidents.
AI Assist	Add agent-driven summaries, recommendation review, and notification routing.
Scale	Replace local JSON persistence with managed database, audit logs, SSO, and BI export pipeline.

Risks and Dependencies

Risk or Dependency	Mitigation
Score model may not match stakeholder judgment.	Calibrate thresholds with historical project outcomes and manager review.
External systems may have inconsistent data quality.	Start with read-only integration pilot and data quality dashboard.
Client-visible data may expose internal details.	Maintain separate internal risks and clientRisks fields with role-scoped rendering.
Alert fatigue may reduce adoption.	Define severity thresholds, digest cadence, and acknowledgement rules.
Local JSON storage is not production-grade.	Use it for demo/MVP only; plan migration to managed database.

8. Agent Graph

Agent Graph

[object Object]

Agent Responsibilities

Agent	Responsibility	Guardrail
Data Ingestion Agent	Collect project, support, CRM, delivery, and feedback signals.	Read-only connectors during pilot.
Signal Normalization Agent	Map imported signals into project, risk, issue, timeline, and sentiment fields.	Preserve source metadata and timestamps.
Health Scoring Agent	Calculate deterministic health score and confidence trend.	Use transparent formula and versioned weights.
Risk Detection Agent	Identify critical, elevated, and watchlist signals.	Separate detected signals from approved escalations.
Score Explanation Agent	Explain why a score changed and which drivers contributed most.	Avoid unsupported claims; cite source signals.
Recovery Recommendation Agent	Propose owner, action, urgency, and follow-up cadence.	Require Project Manager/Admin approval before action.
Notification Agent	Route approved alerts to dashboard, email, Slack, or Teams.	Respect role scope and notification preferences.
Report Generation Agent	Produce scoped executive and delivery summaries.	Exclude client-restricted internal notes.
Human Review	Validate recommendations, acknowledge risk, and approve stakeholder communication.	Required for escalation and client-facing updates.

Discovery Decisions Before Agent Build

1. Confirm authoritative sources for project status, support severity, client sentiment, and delivery milestones.
2. Define score thresholds for Contained, Elevated, Critical, and executive escalation.
3. Define what content is allowed in client-facing summaries.
4. Select first notification channel and escalation owner rules.
5. Decide audit requirements for agent-generated recommendations and human approvals.

Prepared as part of the Customer Health discovery and solution design process.